FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

MATHEMATICS

Past Paper 2024 • Paper Code: 2HA-I 24008 — 40081 (Version B) / 42081 (Version D)

Retyped faithfully from the original board paper — same questions, no changes.

VERSION B

MATHEMATICS HSSC-II — SECTION – A (Objective Type)

Time Allowed: 25 Minutes | **Maximum Marks:** 20 | **Paper Code:** 40081 (Version B)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. If f(x) = √(x+5) and f(g(x)) = 3x then what is the value of g(x)?

(A) 9x^2 + 5
(B) 3x^2 - 5
(C) 9x^2 - 5
(D) 3x^2 + 5

Q2. Without finding the inverse, what is the range of f^(-1)(x), where f(x) = 3 + √(x-2)?

(A) [2,∞)
(B) [3,∞)
(C) [0,∞)
(D) [-2,∞)

Q3. Which of the following represents an even function?

(A) f(x) = (x^2-16)/(x-4)
(B) f(x) = Cos x - x/2
(C) f(x) = (1-Cos x)/Sin^2 x
(D) f(x) = (x-5)^4

Q4. If x + y = Sin(x+y) then dy/dx is equal to:

(A) Cos(x+y)
(B) Sin(x+y)/(x+y)
(C) -1
(D) 1

Q5. Minimum value of the function f(x) = x^2 + 2x - 3 is:

(A) 1
(B) -1
(C) -4
(D) -3

Q6. If f(x) = e^(ln(Sin x)) then what is the value of f'(π/2)?

(A) -1
(B) 0
(C) 1
(D) ∞

Q7. What evaluates ∫(0 to π/2) Cos x (e^(Sin x) dx)?

(A) 2e
(B) e
(C) e+1
(D) e-1

Q8. Which of the following options represents f'(x) = x/(x^2+1) and f(0) = 1?

(A) f(x) = 2ln(x^2+1)+2
(B) f(x) = ln(x^2+1)+1
(C) f(x) = (1/2)ln(x+1)+1
(D) f(x) = (1/2)ln(x^2+1)+1

Q9. If 2∫(0 to 2)(x-k)dx = 1 then what is the value of k?

(A) 3/4
(B) 4/3
(C) 1/2
(D) 3/2

Q10. For what value of k, the line πx + √2 y + √7 = 0 is perpendicular to the line kx + 3y + 2 = 0?

(A) 3/π
(B) 3√2/π
(C) -3√2/π
(D) √2/π

Q11. A line that cuts the x-axis at (2,0) and y-axis at (0,-4), has equation:

(A) 2x + y - 4 = 0
(B) 2x - y - 4 = 0
(C) 2x + y + 4 = 0
(D) x + 2y + 4 = 0

Q12. What is the angle between pair of lines represented by x^2 + 2xy - y^2 = 0?

(A) π/3
(B) π/4
(C) π/6
(D) π/2

Q13. If the lines 3x - y - 2 = 0, 5x + ay - 3 = 0 and 2x + y - 2 = 0 are concurrent then what is the value of a?

(A) -1
(B) -2
(C) -3
(D) -4

Q14. The ordered pair (3,2) is NOT a solution of inequality:

(A) x + y > 2 (B) 3x + 5y > 7

(C) 3x - 7y < 3 (D) x - y > 1

Q15. What are the foci of an ellipse x^2/25 + y^2/16 = 1?

(A) (±4,0) (B) (±3,0)

(C) (0,±3) (D) (±5,0)

Q16. Directrices of the hyperbola x^2/a^2 - y^2/b^2 = 1 are:

(A) x = 0 (B) y = 0

(C) x = ±a/e (D) y = ±e/a

Q17. What is the length of tangent from (2,1) to the circle 3x^2 + 3y^2 + 6x - 12y + 1 = 0?

(A) 4/√3 (B) 2/√3

(C) 4 (D) 2

Q18. If |Sin α i + Cos α j + t k| = 3 then, what is the value of t?

(A) 2 (B) 9

(C) 2√2 (D) 4

Q19. What is Cosine of the angle that a vector v = √2i + j + k makes with y-axis?

(A) 60° (B) 45°

(C) 30° (D) 90°

Q20. What are the direction Cosines of the vector √3i - √3j + √3k?

(A) 1/√3, -1/√3, 1/√3 (B) 1,1,-1

(C) 1,-1,1 (D) √3,-√3,√3

MATHEMATICS HSSC-II — SECTIONS B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 80 | **Paper Code:** 2HA-I 24008 (B)

Note: SECTION – B (Marks 48) — Q.2 Solve the following Questions. (Use of graph paper is not necessary. Candidates can make their own grid on answer book) (12 x 4 = 48)

(i) [04] If f(x) = (3x+2)/(2x-1), find f^(-1)(x) and also show that f^(-1)(f(x)) = x — OR (04) — Find dy/dx if 2y^3 - 3xy^4 + 2x^2y + 5x = 6. Also find the value of dy/dx at (1,1).

(ii) [04] Find the derivative of y = (2√x+2)(x-√x) — OR (04) — Discuss the continuity of the function at x=1: f(x) = {3x-1 if x<1; 4 if x=1; 2x if x>1}.

(iii) [04] If y = Cot(q Cot^(-1) x) then show that (1+x^2)y(1) - q(1+y^2) = 0 — OR (04) — Evaluate lim(θ→0) (Secθ-1)/(θSecθ).

(iv) [04] Examine the function x^3 - 6x^2 + 9x + 3 for extreme values. — OR (04) — Use the differential to approximate value of Sin61°.

(v) [04] Find the area bonded by the curve y = x^3 - 9x and the x-axis. — OR (04) — Find the area of the region bonded by 10x^2 - xy - 21y^2 = 0 and x + y + 1 = 0.

(vi) [04] Solve the differential equation dy/dx = (3/4)x^3 + x - 3 if y=0, when x=2 — OR (04) — Find the point P on the joint of A(1,4) and B(5,6) that is twice as far from A as B is from A and lies on opposite side of A as B does.

(vii) [04] The length of perpendicular form origin to the line is 8 unit and angle of inclination is 30°. Find the slope and y-intercept of the line. — OR (04) — Integrate ∫ (x Sin^(-1) x)/√(1-x^2) dx.

(viii) [04] Find the equation of a circle passing through the point (-2,-5) and touching the line 3x+4y-24=0 at the point (4,3) — OR (04) — Graph the feasible region of the following system of linear inequalities by shading and find the corner points: 3x+2y≥6, x+y≤4, x≥0, y≥0.

(ix) [04] Graph the feasible region of the following system of linear inequalities by shading and find the corner points. 5x+7y≤35, x-2y≤4, x≥0, y≥0 — OR (04) — Find the equation of parabola having Focus (-3,4) and directrix 3x+2y-3=0.

(x) [04] Prove that altitudes of a triangle are concurrent (by vector method). — OR (04) — Find the value of C, when the line 5x+2y+C=0 will touch the hyperbola x^2/4 - y^2/9 = 1.

(xi) [04] Find the points of intersection to the two conics x^2/18 + y^2/8 = 1 and x^2/3 - y^2/3 = 1 — OR (04) — If u = 2i+3j+4k, v = i+4j+3k and w = i+7j+λk represent the side of a triangle. Find the value of λ.

(xii) [04] Find the equation of an ellipse with foci (±√5, 0) and passing through the point (3/2, √3) — OR (04) — Find the moment about (1,1,1) of each of the concurrent forces i-2j, 3i+2j-k and 5j+2k where P(2,0,1) is the point of concurrency.

> Note: SECTION – C (Marks 32) — Note: Solve the following Questions. (Use of graph paper is not necessary. Candidates can make their own grid on answer book) (4 x 8 = 32)

Q3. (Marks 08) Q.3 Let f(x) = {px+2 if 0≤x<2; 7-qx if 2≤x<4; 2x+1 if 4≤x<6}. Find the values of p and q such that f(x) is continuous at x=2, and x=4, sketch the graph of f(x) after finding the values of p and q. — OR (08) — A box with a square base and open top is to have a volume 32 cubic dm. Find the dimensions of the box that will require the least material.

Q4. (Marks 08) Q.4 Evaluate ∫ (2x^2 - x - 7)/((x+2)^2 (x^2+2x+5)) dx — OR (08) — Find the equation of tangent of an ellipse x^2/128 + y^2/18 = 1 which are parallel to the line 3x+8y+1=0. Also find the point of contact.

Q5. (Marks 08) Q.5 The diagram shows a triangle ΔABC where A(-3,-4), B(4,6) and C(4,-3). (A coordinate diagram of triangle ABC is printed in the source scan.) a.

Write the equation of sides AB and AC. b. Find interior angle, ∠A. c. Find the area of ΔABC d. Find the perpendicular distance from point C(4,-3) to line AB — OR (08) — A factory produces two items ceiling lights and ceiling fans by using two machines A and B. Machine A has at most 120 hours available and machine B has maximum 144 hours available. Manufacturing of a ceiling light requires 5 hours in machine A and 4 hours in machine B, while manufacturing of a ceiling fan requires 4 hours in machine A and 8 hours in machine B. If the factory gets a profit of Rs 50 per ceiling light and Rs 80 per ceiling fan, how many ceiling lights and ceiling fans should be manufactured to get maximum profit?

Q6. (Marks 08) Q.6 Find the centre, foci, eccentricity, vertices and equation of directrices of the conic 9x^2 - y^2 - 12x - 2y + 2 = 0 — OR (08) — Evaluate ∫(π/6 to π/4) Cos^2θ Cot^2θ dθ.

VERSION D

MATHEMATICS HSSC-II — SECTION – A (Objective Type)

Time Allowed: 25 Minutes | **Maximum Marks:** 20 | **Paper Code:** 42081 (Version D)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. What result occurs, in evaluating lim(h→0)(1+2h)^(1/h)?

(A) e
(B) 1/e
(C) e^2
(D) 1/e^2

Q2. If f(x) = 3x - 5 then what is the value of f^(-1)(4)?

(A) 2
(B) 3
(C) 1/3
(D) 5/3

Q3. If f(x) = 3x + 1 and f(g(x)) = 3x^2 - 3x + 1 then, what is the value of g(x)?

(A) x^2 - x
(B) x^2 - 3x + 1
(C) 3x^2 - 3x + 3
(D) 3x^2 + x

Q4. If y = √(Sin√x) then dy/dx is equal to:

(A) 1/(4x tan√x)
(B) 4x√(Sin√x)
(C) (Cos√x)/(4√x Sin√x)
(D) (Cos√x)/(4√(Sin√x))

Q5. What is the value of x at the point on the curve y = x^2 - 8x + 3, where the gradient is 2?

(A) -3
(B) 1
(C) -5
(D) 5

Q6. If f(x) = 2Sin^(-1)(x/3) then what is the result of f'(1)?

(A) 3/(2√2)
(B) 1/(2√2)
(C) 1/√2
(D) 3/√2

Q7. If ∫(0 to 2) f(x)dx = 3 then, what is the value of K if ∫(0 to 2)(3f(x)+4x)dx = K?

(A) 20
(B) 17
(C) 11
(D) 1

Q8. If f'(x) = 1/x + 1/(x^2+1) and f(1) = π/2 then f(x) is equal to:

(A) ln|x| + tan^(-1)x + π/2
(B) ln|x| + tan^(-1)x + π/4
(C) ln|x| + cot^(-1)x + π/2
(D) 2ln|x| + cot^(-1)x + π/4

Q9. What result occurs, in evaluating ∫(Sin x - Cos x)/√(1-Sin 2x) dx?

(A) Sin x + c
(B) Sin x - Cos x + c
(C) x + c
(D) Sin x + Cos x + c

Q10. For what value of k the lines 4x - y + k = 0 and 6kx + 8y + 3 = 0 are perpendicular?

(A) 1/2
(B) 1/3
(C) 1/4
(D) 1/6

Q11. The perpendicular distance from origin to a line 4x + 3y + 10 = 0 is:

(A) 0
(B) 1
(C) 2
(D) 10

Q12. Centroid of the triangle with vertices A(4,-2), B(-2,4) and C(4,10) is:

(A) (-2,4)
(B) (2,4)
(C) (2,-4)
(D) (-2,-4)

Q13. The ordered pair (1,2) is one solution of the inequality:

(A) 2x + y > 5
(B) 2x - y > 3
(C) 2x + y < 3
(D) 2x + y < 5

Q14. What is the length of diameter of a circle (x+5)^2 + (y-8)^2 = 12?

(A) 4√3 (B) 2√3

(C) 12 (D) 24

Q15. What is the length of latus rectum of an ellipse x^2/36 + y^2/25 = 1?

(A) 25/6 (B) 25/36

(C) 25/3 (D) 3/25

Q16. Which of the following is an asymptote of the hyperbola 3x^2 - 4y^2 - 12 = 0?

(A) y = ±(4/3)x (B) y = ±(3/4)x

(C) y = ±(2/√3)x (D) y = ±(√3/2)x

Q17. The directrix of the parabola x^2 = 8y is:

(A) x + 2 = 0 (B) x - 2 = 0

(C) y + 2 = 0 (D) y - 2 = 0

Q18. If a = i + 2j + k, b = j + k then what is the value of λ if a + λb is perpendicular to b?

(A) -3/2 (B) -1/2

(C) 1/2 (D) 3/2

Q19. What is Cosine of the angle that a vector √2i + j + k makes with x-axis?

(A) 30° (B) 45°

(C) 60° (D) 90°

Q20. If the vectors 2i - 3j + 4k, i + 2j - k and αi - j + 2k are coplanar then, what is the value of α?

(A) 8/11 (B) 11/8

(C) 5/8 (D) 5/11

MATHEMATICS HSSC-II — SECTIONS B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 80 | **Paper Code:** 2HA-I 24008 (D)

Note: SECTION – B (Marks 48) — Q.2 Solve the following Questions. (Use of graph paper is not necessary. Candidates can make their own grid on answer book) (12 x 4 = 48)

(i) [04] Given that f(x) = x^3 - px^2 + qx + 1 if f(3) = -5 and f(-2) = 5 then find the values of p and q — OR (04) — Show that dy/dx = y/x if x/y = tan^(-1)(x/y).

(ii) [04] Find the derivative of x^(2/3) by using definition w.r.t. x. Also calculate the derivative at x=16 — OR (04) — For the given function f(x) = {2ax+3 if x<2; 2x+2a if x≥2}. Find the value of a that lim(x→2) f(x) exists.

(iii) [04] Evaluate ∫(0 to π/2) Cos^3 x √(Sin x) dx — OR (04) — Evaluate the limit lim(θ→0) (1-Cos3θ)/(1-Cos5θ).

(iv) [04] Find the point on the curve y = x^2 + 2 that is closest to the point (18,2) — OR (04) — Evaluate ∫ ln(x + √(x^2+1)) dx (by parts).

(v) [04] Find dy/dx if y = ln[(x(x^2-3)^2)/((x^2-4)^(1/2))] — OR (04) — Find the area above the x-axis bounded by the curve y^2 = 3 - x from x = -1 to x = 2.

(vi) [04] Find the equation of an ellipse with vertices (-1,1), (5,1) and foci (4,1), (0,1) — OR (04) — If two vertices of an equilateral triangle are A(-4,0) and B(4,0). Find the third vertex. How many of these triangles are possible?

(vii) [04] Graph the feasible region subject to the following constraints 2x+8y≤60, 4x+4y≤60, x≥0, y≥0 Also find the corner points. — OR (04) — Find a joint equation of the straight line through the origin represented by 2x^2 - 5xy + 3y^2 = 0.

(viii) [04] Find the equation of a straight line through intersection of the lines x+2y+3=0, 3x+4y+7=0 and making equal intercept on axes. — OR (04) — Graph the feasible region subject to the following constraints 3x-y≥-1, x+y≤5, x≥0, y≥0 Also find the corner points.

(ix) [04] In any triangle ABC, Prove that c^2 = a^2 + b^2 - 2abCosC (Use vector method) — OR (04) — Write the equation of parabola with given elements directrix y=3, vertex (2,2).

(x) [04] Solve the differential equation y - x(dy/dx) = 3(1 + x(dy/dx)). Also find the particular solution if y=7 and x=16 — OR (04) — Find the equation of the normal to the parabola y^2 = 8x which is parallel to the line 2x+3y=10.

(xi) [04] Find the equation of hyperbola with center (2,2) horizontal transverse axis of length 6 and eccentricity e=2 — OR (04) — Find a unit vector perpendicular to the plane containing a and b. Also find the sine of the angle between them, a = 2i-3j+4k, b = -i-j-k.

(xii) [04] Write the equation of tangent line to the circle x^2 + y^2 + 4x + 2y = 0 drawn from (-1,2). Also find the tangential distance. — OR (04) — Find the volume of tetrahedron whose vertices are A(2,1,8), B(3,2,9), C(2,1,4) and D(3,3,10).

> Note: SECTION – C (Marks 32) — Note: Solve the following Questions. (Use of graph paper is not necessary. Candidates can make their own grid on answer book) (4 x 8 = 32)

Q3. (Marks 08) Q.3 Let f(x) = {lx+2m if 0≤x<1; mx+2 if 1≤x<2; 3lx+4 if 2≤x<4}. a. Find the values of l and m such that function f is continuous at x=1 and x=2 b. After finding values of l and m, sketch the graph of f(x) — OR (08) — Discuss the function defined as f(x) = Sin x + (1/(2√2)) Cos x for extreme values in the interval (0,2π).

Q4. (Marks 08) Q.4 Find the centre, foci, eccentricity, vertices and directrices of the conic 9x^2 - 18x + 4y^2 + 8y - 23 = 0 — OR (08) — Show that ∫e^(ax) Sin bx dx = (1/√(a^2+b^2)) Sin(bx - tan^(-1)(b/a)) + c. By using this formula evaluate ∫e^(2x) Sin 3x dx.

Q5. (Marks 08) Q.5 Find the distance between the parallel lines: 12x+5y-6=0, 12x+5y+13=0 a. Sketch the lines. b. Find the equation of parallel lines lying mid way between them. c. Write the equation of line in two intercept from. — OR (08) —

A toy factory manufactures two types of toys: remote car and scooty by using two machines A and B. Machine A has at most 110 hours available and machine B has a maximum of 300 hours available. Manufacturing a remote car requires 5 hours in machine A and 10 hours in machine B. While manufacturing of scooty requires 30 hours in machine A and 20 hours in machine B. If the factory gets profit of Rs. 300 per remote car and Rs. 400 per scooty, how many both toys should be manufactured to get maximum profit?

Q6. (Marks 08) Q.6 Evaluate ∫ (2x^2 - x - 7)/((x+3)(x-2)(x^2-2x+3)) dx — OR (08) — Find the equation of tangent and normal at a point (5, 16/9) to the hyperbola x^2/9 - y^2/16 = 1. Also find the value of c, when the line y = -x + c will touch the hyperbola x^2/16 - y^2/9 = 1.